

RIDHIMA SRIVASTAVA — 365-DAY MASTER STUDY TRACKER

2nd Year B.Tech Computer Science (ABES) | Target: October 2027 Cybersecurity Internship

Daily Schedule: 2 Hours DSA (C++) + 1 Hour Cybersecurity | 3 Hours Total Daily

MONTH 1: C++ BASICS & NETWORK FUNDAMENTALS (DAYS 1 - 30)

Day 1	Video 003: Intro to C++ & Input Operators	OSI Model Overview: All 7 Layers
Day 2	Video 004: Conditionals Part 1 (if/else)	OSI Model: Layers 1 to 4 (Physical to Transport)
Day 3	Video 005: Conditionals Part 2 (Nested & Switch)	TCP vs UDP Protocols & Header structures
Day 4	Video 006: Loops Part 1 (for/while loops)	IP Addressing: IPv4 vs IPv6 & Subnetting
Day 5	Video 007: Loops Part 2 (Factorial & Prime)	MAC Addresses & ARP Protocol
Day 6	Video 008: Pattern Printing Part 1	Port Numbers & Services (21, 22, 80, 443)
Day 7	Video 009: Pattern Printing Part 2	DNS Resolution Process Steps
Day 8	Video 010: Fundamentals Live Session	Wireshark Overview & Interface Setup
Day 9	Video 011: Functions & Pass by Value/Ref	Wireshark Packet Capture: Inspect HTTP
Day 10	Video 012: Pointers & Memory Addresses	Wireshark Display Filters (http, ip.src)
Day 11	Video 013: 1D Arrays Part 1 (Traversal)	Information Security CIA Triad Overview
Day 12	Video 014: 1D Arrays Part 2 (Vectors)	Confidentiality & Symmetric Encryption
Day 13	Video 015: 1D Arrays Part 3 (Two Pointers)	Integrity & Cryptographic Hashing
Day 14	LeetCode #1: Two Sum & #26: Duplicates	Availability & DDoS Attack Vectors
Day 15	LeetCode #121: Best Time to Buy/Sell Stock	TryHackMe Pre-Security Room 1
Day 16	Video 016: 2D Arrays Part 1 (Matrices)	TryHackMe Pre-Security Room 2
Day 17	Video 017: 2D Arrays Part 2 (Transpose)	HTTP Request Methods (GET, POST, PUT)
Day 18	Video 018: 2D Arrays Part 3 (Spiral Order)	HTTP Response Status Codes (200, 404)
Day 19	Video 019: Strings Part 1 (std::string)	Cookies, Sessions & LocalStorage
Day 20	Video 020: Strings Part 2 (Valid Anagrams)	TryHackMe Pre-Security Room 3
Day 21	Video 021: 2D Arrays Live Class	OWASP Top 10 Security Overview
Day 22	Video 022: Time Complexity (Big-O)	OWASP #1: Broken Access Control
Day 23	Video 023: Space Complexity Analysis	OWASP #2: Cryptographic Failures
Day 24	Video 024: Bubble Sorting $O(N^2)$	OWASP #3: Injection Attacks Overview
Day 25	Video 025: Selection & Insertion Sort	SQL Injection (SQLi) Syntax & Logic
Day 26	Video 026: Complexity Live Session	PortSwigger Academy: SQLi Lab 1
Day 27	Video 027: Problems on Sorting	PortSwigger Academy: SQLi Lab 2
Day 28	Video 028: Searching Part 1 (Binary Search)	Cross-Site Scripting (XSS) Overview
Day 29	Video 029: Searching Part 2 (Bounds)	PortSwigger Academy: XSS Lab 1
Day 30	Video 030: Searching Part 3 (Answer Space)	Month 1 Security Review & THM Room

MONTH 2: RECURSION, ADVANCED SORTING & OWASP SECURITY (DAYS 31 - 60)

Day 31	Video 031: Recursion Part 1 (Base Cases)	PortSwigger Academy: XSS Lab 2
Day 32	Video 032: Recursion Part 2 (Fibonacci)	PortSwigger Academy: XSS Lab 3
Day 33	Video 033: Recursion Part 3 (Subsets)	PortSwigger Academy: XSS Lab 4
Day 34	Video 034: Sorting Live Session	Cross-Site Request Forgery (CSRF)
Day 35	Video 035: Recursion Part 4 (Backtracking)	PortSwigger Academy: CSRF Lab 1
Day 36	Video 036: Merge Sort $O(N \log N)$	TryHackMe: Web Fundamentals Room 1
Day 37	Video 037: Quick Sort & Partitioning	TryHackMe: Web Fundamentals Room 2
Day 38	Video 038: Recursion Live Session	TryHackMe: Web Fundamentals Room 3
Day 39	Video 039: Cyclic Sort Pattern	TryHackMe: Web Fundamentals Room 4
Day 40	Video 040: Prefix Sum Algorithm	TryHackMe: Web Fundamentals Room 5
Day 41	Video 041: Sliding Window Technique	TryHackMe: Web Fundamentals Room 6
Day 42	Video 042: Sieve of Eratosthenes	Deloitte Cyber Job Sim Task 1 Review
Day 43	Video 072: Greedy Algorithms Part 2	Deloitte Cyber Job Sim Task 2 Review
Day 44	Video 073: Greedy Algorithms Part 3	Deloitte Cyber Job Sim Task 3 Review
Day 45	Video 074: DP 1 Part 1 (Memoization)	OverTheWire: Bandit Level 0 to 2
Day 46	Video 075: DP 1 Part 2 (Tabulation)	OverTheWire: Bandit Level 3 to 5
Day 47	Video 076: DP 1 Part 3 (1D DP)	OverTheWire: Bandit Level 6 to 8
Day 48	LeetCode #53: Maximum Subarray	OverTheWire: Bandit Level 9 to 10

Day 49	LeetCode #15: 3Sum (Two Pointers)
Day 50	LeetCode #11: Container With Most Water
Day 51	LeetCode #42: Trapping Rain Water
Day 52	LeetCode #3: Longest Substring
Day 53	LeetCode #76: Minimum Window Substring
Day 54	LeetCode #56: Merge Intervals
Day 55	LeetCode #49: Group Anagrams
Day 56	Striver's Sheet: Arrays Medium 1
Day 57	Striver's Sheet: Arrays Medium 2
Day 58	Striver's Sheet: Two Pointers
Day 59	Striver's Sheet: Sliding Window
Day 60	Month 2 DSA Revision & Code Clean Up

[Linux User Management \(useradd\)](#)
[Linux Group Permissions \(chmod\)](#)
[Linux File System Hierarchy \(/var\)](#)
[Shell Scripting: Bash Variables](#)
[Shell Scripting: Log Search script](#)
[Recon Tools: whois, dig, nslookup](#)
[Passive Information Gathering](#)
[Active Information Gathering](#)
[PortSwigger Auth Vulnerabilities 1](#)
[PortSwigger Auth Vulnerabilities 2](#)
[Month 2 Security Review & THM](#)
[Month 2 Cybersecurity Assessment](#)

MONTHS 3 TO 12 DAILY ROADMAP & DIRECT LINKS OVERVIEW

- Month 3 (Days 61-90): Linked Lists, Stacks, Queues (LeetCode) | Linux CLI (OverTheWire Bandit: <https://overthewire.org/wargames/bandit/>)
- Month 4 (Days 91-120): Binary Search Mastery (LeetCode) | Network Recon & Nmap (<https://nmap.org/>)
- Month 5 (Days 121-150): Hash Maps, Trees & Trie (LeetCode) | Cryptography AES/RSA (<https://cryptohack.org/>)
- Month 6 (Days 151-180): Graph BFS/DFS & Shortest Path | SIEM & Splunk Log Analysis (<https://tryhackme.com/path/outline/soclevel1>)
- Month 7 (Days 181-210): Dynamic Programming 1D/2D | Vulnerability Scanners Nessus/OpenVAS (<https://www.openvas.org/>)
- Month 8 (Days 211-240): Advanced LeetCode Sprint | Burp Suite Web Pentesting (<https://portswigger.net/web-security/burp>)
- Month 9 (Days 241-270): Project 1 - C++ Encrypted Vault | OpenSSL C++ API (<https://www.openssl.org/>)
- Month 10 (Days 271-300): Project 2 - C++ Packet Sniffer | Scapy & Raw Sockets (<https://scapy.net/>)
- Month 11 (Days 301-330): LeetCode Top 150 Interview Questions | CompTIA Security+ SY0-701 (<https://www.professormesser.com/>)
- Month 12 (Days 331-365): Resume & Portfolio Launch | October 2027 Internship Hiring Applications (https://github.com/RIDHIMA0120/RIDHIMA_PORTFOLIO)